

THEMATIC RESEARCH REPORT

Quantum Computing & Quantum Chip Manufacturing

The Race to Commercialize the Next Computing Paradigm

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Sector: Technology | Sub-Sector: Quantum Computing Hardware & Software

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1. Executive Summary

Quantum computing stands at a commercial inflection point. The global quantum computing market, valued at approximately \$17.4 billion in 2025 for photonic integrated circuits alone [1], is projected to expand at a 20.8% CAGR through 2034, reaching \$86.4 billion [1]. Boston Consulting Group projects the broader quantum technology market could generate up to \$97 billion in revenue by 2035, with quantum computing capturing \$72 billion of that total [2]. McKinsey has projected the total addressable market at \$850 billion over the coming decade [3].

Key findings of this report:

- IBM and Alphabet (Google) dominate quantum computing patent filings, accounting for the majority of the 300+ quantum computing patents issued in 2025 [4]. D-Wave holds the third-largest global portfolio with 260+ US granted patents and 550+ worldwide [5].
- The industry is transitioning from proof-of-concept to commercialization, with IBM targeting quantum advantage by 2026 and fault-tolerant systems by 2029 [6]. Google achieved a landmark error-correction milestone with its 105-qubit Willow chip [7].
- Pure-play quantum stocks remain pre-profit, with revenue measured in single-digit millions quarterly. IonQ leads on revenue (\$64.7M in Q1 2026), followed by D-Wave (\$2.9M) and Rigetti (\$4.4M) [8][9][10].
- Government investment is accelerating globally, with China estimated to invest \$2.5-3.0 billion annually and the US awaiting NQI reauthorization [11]. The first three quarters of 2025 attracted \$3.77 billion in private quantum funding, a 128% year-over-year surge [2].
- Quantinuum, the Honeywell spin-off, is preparing for a Nasdaq IPO led by J.P. Morgan and Morgan Stanley, aiming to deliver the first commercial-scale, fully fault-tolerant quantum computer by decade-end [12].

2. Theme Overview: Quantum Computing

What is Quantum Computing? Quantum computing leverages quantum mechanical phenomena, specifically superposition and entanglement, to process information in ways fundamentally different from classical computers. While classical bits exist in states of 0 or 1, quantum bits (qubits) can exist in multiple states simultaneously, enabling exponential speedups on certain problem classes including molecular simulation, optimization, cryptography, and machine learning.

Why It Matters Now. After years of laboratory demonstration, quantum computing is gaining genuine commercial momentum. NatWest Group's 2026 technology outlook noted that "quantum computing is now gaining genuine commercial momentum, with substantial investment focused on enhancing computational power, reducing error rates, and tackling a range of practical challenges" [13]. The convergence of improved error correction, increasing qubit counts, and expanding cloud-based access models is accelerating the path toward practical, enterprise-grade quantum systems. KB Securities noted that "the quantum sector reached a commercial inflection point" in 2025-2026 [6].

Commercialization Timeline. The consensus commercialization timeline of 2029-2030 is being expedited by advancements in hybrid architectures and error correction technologies [14]. The cloud-based quantum computing market alone is projected to grow to approximately \$4 billion by 2028, at a 38.5% CAGR [14]. Key milestones include IBM targeting practical industrial algorithms capable of 7,500 gate operations in 2026 and a fully fault-tolerant system by 2029, while Google aims for practical commercial applications within five years [6].

3. Market Size and Growth Projections

Market Segment	2023 Value	2025E Value	2030E/2034E Value	CAGR	Source
Photonic IC Market	\$4.7B	\$17.4B	\$86.4B (2034)	20.8%	Fortune Business Insights [1]
Optical QC Platform	\$4.7B	-	\$29.6B (2030)	30.5%	Valuates Reports [15]
Total Quantum Tech (BCG)	-	-	\$97B (2035)	-	BCG Global [2]
QC Computing (BCG)	-	-	\$72B (2035)	-	BCG Global [2]
QC Optimization (BCG)	-	-	\$100-220B	-	D-Wave/BCG [9]
Cloud-based QC	-	-	\$4B (2028)	38.5%	KB Securities [14]
Total QC TAM (McKinsey)	-	-	\$850B	-	PatentVest/McKinsey [3]

Global government and private investment is surging. China is estimated to invest \$2.5-3.0 billion annually in quantum computing, primarily in superconducting technologies [11]. The first three quarters of 2025 attracted \$3.77 billion in private quantum funding, representing a 128% year-over-year increase [2]. The UN designated 2025 as the International Year of Quantum Science and Technology, signaling global institutional alignment with the sector [2].

4. Technology Landscape: Qubit Modalities

The quantum computing industry encompasses multiple competing hardware approaches, each with distinct advantages and challenges:

Qubit Type	Key Players	Advantages	Challenges	Maturity
Superconducting	IBM, Google, Rigetti, Origin Quantum	Fast gate speeds, mature fabrication	Requires extreme cooling (<20 mK)	Most advanced
Trapped Ion	IonQ, Quantinuum, Alpine Computing	High fidelity, long coherence times	Slower gate speeds, scaling complexity	Advanced
Photonic	Xanadu, QCI, PsiQuantum	Room temperature, networkable	Photon loss, deterministic gates	Emerging
Topological	Microsoft (Majorana 1)	Inherent error protection	Still in early development	Early stage
Neutral Atom	Infleqtion, QuEra, Atom Computing	Scalable arrays, flexible connectivity	Relatively new approach	Emerging
Quantum Annealing	D-Wave	Commercial today for optimization	Limited to optimization problems	Commercial

Recent Chip Milestones: Google's 105-qubit Willow chip demonstrated exponential error reduction, going 'below threshold' for the first time, a landmark for fault-tolerant computing [7]. IBM unveiled its Starling roadmap targeting 200 logical qubits by 2029. Microsoft introduced the Majorana 1 topological qubit architecture, aiming to scale to a million qubits [2]. China's Jiuzhang 4.0 manipulated up to 3,050 photons, a leap from 255 in the previous generation [16].

5. Competitive Landscape: Public Companies

5.1 Pure-Play Quantum Computing Companies

Company	Ticker	Mkt Cap	Qubit Type	Key Focus	Revenue (LTQ)
IonQ	IONQ	\$21.5B	Trapped Ion	Full-stack QC systems & networking	\$64.7M
D-Wave Quantum	QBITS	\$8.1B	Annealing + Gate	Commercial optimization solutions	\$2.9M
Rigetti Computing	RGTI	\$6.4B	Superconducting	Full-stack QPU design & fab	\$4.4M
Quantum Computing Inc.	QUBT	\$1.6B	Photonic	Integrated photonics & sensing	\$3.7M
Infleqtion	INFQ	Newly Listed	Neutral Atom	QC, sensing & software	\$9.5M
Xanadu QT	XNDU	Newly Listed	Photonic	Networked modular QC (Aurora)	Pre-revenue
Horizon Quantum	Recent IPO	New	Agnostic (SW)	Hardware-agnostic QC software	Pre-revenue

5.2 Large-Cap Companies with Quantum Divisions

Company	Ticker	Mkt Cap	Quantum Initiative	Key Chip/System
Alphabet (Google)	GOOGL	~\$2.3T	Google Quantum AI Lab	Willow (105-qubit superconducting)
IBM	IBM	~\$260B	IBM Quantum	Starling roadmap, Heron processor
Microsoft	MSFT	~\$3.3T	Azure Quantum	Majorana 1 (topological)
Amazon	AMZN	~\$2.2T	AWS Braket / Ocelot	Cat qubit error correction
Honeywell	HON	~\$155B	Majority owner of Quantinuum	H-Series trapped-ion systems

6. Competitive Landscape: Private Companies

Company	HQ	Qubit Type	Key Focus	Notable
Quantinuum	UK/US	Trapped Ion	Full-stack fault-tolerant QC	IPO imminent; Honeywell spin-off [12]
PsiQuantum	US	Photonic	Million-qubit photonic QC	\$700M+ raised; GlobalFoundries fab partner
QuEra Computing	US	Neutral Atom	Error-corrected neutral-atom QC	\$230M from Google & SoftBank [7]
Atom Computing	US	Neutral Atom	Scalable neutral-atom arrays	Demonstrated 1,225 atomic qubits
Alpine Computing	US	Trapped Ion	Trapped-ion QC	Quantinuum peer company [17]
Origin Quantum	China	Superconducting	Full-stack QC + OS	Concentrated patent portfolio [3]
Alice & Bob	France	Cat Qubit	Error-corrected cat qubits	Hardware-efficient error correction
NuCrypt	US	Photonic	Quantum communications	Acquired by QCi in March 2026 [18]

Quantinuum IPO Watch. Honeywell's quantum computing spin-off Quantinuum has filed an S-1 with the SEC and is preparing for a Nasdaq IPO with J.P. Morgan and Morgan Stanley as lead bookrunners [12]. The company aims to deliver the first commercial-scale, fully fault-tolerant quantum computer by decade-end. Quantinuum identifies Alphabet, IBM, QuEra, IonQ, Atom Computing, Alpine Computing, and Rigetti as its peer companies [17].

7. Patent and Intellectual Property Analysis

Intellectual property is a critical competitive moat in quantum computing. PatentVest noted that "leadership in quantum computing will not be decided by scientific milestones or funding size alone. Defensible intellectual property, coupled with the ability to commercialize, will define the winners in this \$850B market" [3].

7.1 Patent Portfolio Rankings

Company	Type	US Patents Granted	Global (Granted+Pending)	Key IP Domains
IBM	Public	Leader (~100+ in 2025 alone)	Largest globally	Superconducting circuits, error correction, middleware [4]
Alphabet (Google)	Public	#2 (~large share of 300 in 2025)	Among top globally	Quantum hardware, algorithms, error correction [4]
D-Wave Quantum	Public	260+	550+ (granted + pending)	Annealing, gate-model, on-chip control, cryogenic [5]
IonQ	Public	~112 issued	~400 (granted + pending, incl. networking)	Trapped-ion, quantum networking [19]
Rigetti Computing	Public	~112 issued	~252 (issued + pending)	Superconducting QPU design, fabrication [20]
Intel	Public	Top 5 globally	Significant	Spin qubits, silicon-based quantum [3]
Microsoft	Public	Top 5 globally	Significant	Topological qubits, quantum software [3]
Origin Quantum	Private	Concentrated portfolio	Targeted	Superconducting QC (China) [3]
QCi (with LSI/NuCrypt)	Public	~25+ (Freedom Photonics alone)	Growing via M&A	Photonics, TFLN, quantum optics [18][21]
Fujitsu	Public	Top 10 globally	Significant	QC technologies, hybrid HPC-quantum [22]

Key observations: Of the approximately 300 quantum computing patents issued in 2025, IBM and Google received the vast majority, with a steep drop-off to other players [4]. Financial institutions including Wells Fargo, Bank of America, and JPMorgan also hold quantum patents, reflecting exploratory "big company spend" [4]. The Japan Patent Office ranked the top global quantum computing patent applicants (2010-2021) as IBM, Origin Quantum, Microsoft, Intel, Google, D-Wave, Northrop Grumman, Beijing Baidu, Fujitsu, and Rigetti [22].

8. Public Company Fundamental Analysis (Last 6 Quarters)

8.1 IonQ, Inc. (IONQ) - NYSE

Market Cap: \$21.5B | Price: \$57.47 | CEO: Niccolo de Masi

Analyst Consensus: 3 Buy, 3 Hold, 0 Sell [8]

Quarter	Revenue (\$M)	Op. Income (\$M)	Net Income (\$M)	EPS (Diluted)
Q1 2026	64.7	(271.5)	805.4	\$2.07
Q4 2025	61.9	(242.7)	703.5	\$1.93
Q3 2025	39.9	(168.8)	(1,055.0)	(\$3.58)
Q2 2025	20.7	(146.5)	(176.8)	(\$0.70)
Q1 2025	7.6	(75.7)	(32.3)	(\$0.14)
Q4 2024	11.7	(76.0)	(202.0)	(\$0.93)

Quarter	Total Assets (\$M)	Cash + STI (\$M)	Total Debt (\$M)	Equity (\$M)
Q1 2026	6,694	2,033	30	4,976
Q4 2025	6,570	2,392	30	3,800
Q3 2025	4,319	1,082	29	2,273
Q2 2025	1,347	543	19	1,162
Q1 2025	850	588	17	765
Q4 2024	508	340	18	384

Commentary: IonQ has delivered the strongest revenue trajectory among pure-play quantum companies, growing from \$7.6M in Q1 2025 to \$64.7M in Q1 2026, an over 8x year-over-year increase. The company holds \$2.0B in cash and short-term investments against minimal debt (\$30M), providing a substantial runway. However, operating cash flow remains deeply negative at (\$151M) in Q1 2026, reflecting continued heavy R&D investment. The large positive net income figures in Q4 2025 and Q1 2026 are primarily driven by non-cash items related to equity fair-value adjustments, not operational profitability [8].

8.2 Rigetti Computing, Inc. (RGTI) - NASDAQ

Market Cap: \$6.4B | Price: \$19.27 | CEO: Subodh Kulkarni

Analyst Consensus: 6 Buy, 1 Hold, 0 Sell [10]

Quarter	Revenue (\$M)	Op. Income (\$M)	Net Income (\$M)	EPS (Diluted)
Q1 2026	4.4	(26.0)	33.1	(\$0.06)
Q4 2025	1.9	(22.6)	(18.2)	(\$0.06)
Q3 2025	1.9	(20.5)	(201.0)	(\$0.62)
Q2 2025	1.8	(19.9)	(39.7)	(\$0.13)
Q1 2025	1.5	(21.6)	42.6	\$0.15
Q4 2024	2.3	(18.5)	(153.0)	(\$0.80)

Commentary: Rigetti operates a vertically integrated quantum computing model with in-house QPU fabrication, a differentiator in the space. Revenue remains modest at \$4.4M in Q1 2026. The company has \$48M in cash (down from \$444M in Q4 2025, likely reflecting capital deployment for fab expansion), minimal debt (\$6.8M), and \$584M in equity. Operating cash burn is approximately (\$16M) per quarter. The company holds 252 issued and pending patents [10][20].

8.3 D-Wave Quantum Inc. (QBTS) - NYSE

Market Cap: \$8.1B | Price: \$22.15 | CEO: Alan Baratz

Analyst Consensus: 13 Buy, 0 Hold, 0 Sell (unanimous Buy) [9]

Quarter	Revenue (\$M)	Op. Income (\$M)	Net Income (\$M)	EPS (Diluted)
Q1 2026	2.9	(54.7)	(18.4)	(\$0.05)
Q4 2025	2.8	(34.8)	(42.3)	(\$0.12)
Q3 2025	3.7	(27.7)	(140.0)	(\$0.41)
Q2 2025	3.1	(26.5)	(167.3)	(\$0.57)
Q1 2025	15.0	(11.3)	(5.4)	(\$0.02)
Q4 2024	2.3	(20.2)	(86.1)	(\$0.37)

Commentary: D-Wave is the only company offering commercial quantum annealing systems and claims the quantum computing optimization market is worth \$100-220 billion [9]. The company expanded its competitive scope in January 2026 through the acquisition of Quantum Circuits, gaining dual-rail qubit gate-model technology. D-Wave holds \$588M in cash with minimal debt, and 550+ global patents. Revenue of \$15M in Q1 2025 was anomalously high (likely a large contract); normalized quarterly revenue is approximately \$3M. The unanimous 13-0 Buy analyst rating is the strongest consensus in the sector [9].

8.4 Quantum Computing, Inc. (QUBT) - NASDAQ

Market Cap: \$1.6B | Price: \$11.73 | CEO: Yuping Huang

Analyst Consensus: 3 Buy, 1 Hold, 0 Sell [21]

Quarter	Revenue (\$M)	Op. Income (\$M)	Net Income (\$M)	EPS (Diluted)
Q1 2026	3.7	(20.6)	(4.1)	(\$0.02)
Q4 2025	0.2	(22.2)	(1.6)	(\$0.01)
Q3 2025	0.4	(10.4)	2.4	\$0.01
Q2 2025	0.1	(10.2)	(36.5)	(\$0.26)
Q1 2025	0.04	(8.3)	17.0	\$0.11
Q4 2024	0.06	(8.9)	(51.2)	(\$0.55)

Commentary: QCi completed two transformative acquisitions in Q1 2026: Luminar Semiconductor (LSI), bringing photonic components manufacturing and ~25 patents, and NuCrypt, adding quantum communications IP spanning quantum optics and RF photonics [18][21]. Revenue jumped to \$3.7M in Q1 2026 from near-zero levels, primarily driven by LSI's existing customer base. The company holds \$986M in cash with near-zero debt, supported by \$1.6B in equity after significant capital raises. QCi differentiates through room-temperature, low-power photonic quantum systems and an in-house TFLN foundry for chip fabrication [21].

9. Sector Impact Matrix

End Market	Impact Level	Direction	Timeline	Key Use Cases
Pharmaceuticals / Drug Discovery	Very High	Positive	Medium-term	Molecular simulation, catalyst design
Financial Services	High	Positive	Near-term	Portfolio optimization, risk modeling, fraud
Cybersecurity / PQC	Very High	Mixed	Near-term	Post-quantum crypto migration (urgency)
Defense / National Security	Very High	Positive	Near-term	Sensing, secure comms, optimization
Materials Science	High	Positive	Medium-term	Battery materials, advanced alloys
Logistics / Supply Chain	High	Positive	Medium-term	Route optimization, scheduling
Telecommunications	Medium	Positive	Long-term	Quantum networking, secure infrastructure
Energy / Utilities	Medium	Positive	Long-term	Grid optimization, materials for renewables
Artificial Intelligence	High	Positive	Medium-term	Quantum-enhanced ML, hybrid QC-AI

Quantum-AI convergence is an emerging theme. Inflection's collaboration with NVIDIA on quantum processor calibration using AI models exemplifies the hybrid quantum-AI approach gaining traction [23]. IBM demonstrated a quantum-assisted protein simulation with 210x accuracy improvement, underscoring near-term pharma applications [24].

10. Investment Playbook

Direct Beneficiaries (Pure-Play Exposure)

Company	Ticker	Exposure Type	Investment Thesis
IonQ	IONQ	Trapped-ion leader	Fastest revenue growth; ~400 quantum networking patents; \$2B+ cash runway
D-Wave Quantum	QBITS	Only commercial QC provider	Sole commercial quantum annealer; 550+ patents; now also gate-model via QCI acq.
Rigetti Computing	RGTI	Vertically integrated	In-house QPU fab; strong analyst consensus (6-1); superconducting leader among startups
Quantum Computing Inc.	QUBT	Photonic innovator	Room-temperature photonic approach; integrated manufacturing via LSI acquisition
Infleqtion	INFQ	Neutral-atom pioneer	First publicly traded neutral-atom QC company; defense/govt. contracts

Large-Cap Quantum Exposure

For investors seeking quantum exposure with lower risk, Alphabet (GOOGL), IBM, and Microsoft offer quantum computing programs within diversified, profitable businesses. Alphabet and IBM dominate the patent landscape and are investing tens of billions in quantum R&D [4]. Honeywell (HON) provides indirect exposure through its majority stake in Quantinuum, expected to IPO imminently [12].

Upcoming Catalysts

Timeline	Catalyst	Companies Affected
H1 2026	Quantinuum Nasdaq IPO	HON, IONQ, RGTI (comp. valuation)
2026	IBM 7,500 gate-operation milestone	IBM, all QC pure-plays
2026	US NQI Reauthorization Act	Sector-wide (funding)
2026-2027	Google quantum commercial applications	GOOGL, sector
2027-2029	Fault-tolerant system demonstrations	IBM, Quantinuum, Google
2029-2030	Commercial quantum advantage	All quantum players

11. Key Risks

Risk Category	Description	Likelihood	Impact
Technology Risk	Failure to achieve scalable fault-tolerant quantum computing within projected timelines	Medium	Very High
Funding / Cash Burn	Pure-play companies burn \$15-150M/quarter; most are years from profitability	High	High
Valuation Risk	Current market caps (\$1.6B-\$21.5B) are far above fundamental value for pre-profit companies	High	High
Competition / Crowding	Multiple qubit modalities competing; unclear which approach will dominate at scale	Medium	Medium
Government Policy	Delays in NQI reauthorization; geopolitical restrictions on quantum technology exports	Medium	Medium
Dilution Risk	Continued equity issuance to fund operations erodes existing shareholder value	High	Medium
Talent Scarcity	Limited global pool of quantum hardware engineers and physicists	High	Medium
Quantum-Safe Crypto Transition	If quantum crypto-breaking arrives faster than expected, disruption to financial systems	Low-Med	Very High

12. Sources

#	Source	Date	URL
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2	PartnerCap Securities - SPAC Monthly Monitor	Apr 2026	https://app.bigdata.com/documents/DA318F369935CBB8F27E7E65FB13705B
3	PatentVest - Quantum Computing Rankings (GlobeNewswire)	Aug 2025	https://app.bigdata.com/documents/B612F27225CDAB5943AC90B0C2B790BE
4	Benzinga - IBM, Google Emerge As Top Investors in QC Patents	Jun 2025	https://app.bigdata.com/documents/26F5C450908B8965B12D9DA228F90F47
5	D-Wave Q4 2024 Earnings Presentation	Mar 2025	https://app.bigdata.com/documents/0F439BD4E2D0AD9091F364589A2F0823
6	KB Securities - World Quantum Day Report	Apr 2026	https://app.bigdata.com/documents/82E1942F74DCFCA37058A080AE7D249E
7	Alphabet Q4 2025 Earnings Call Report	Feb 2026	https://app.bigdata.com/documents/E247BFCE6C73239B559858EB8275E01C
8	IonQ Company Tearsheet (Bigdata.com)	May 2026	https://bigdata.com
9	D-Wave Q1 2026 Earnings Call Transcript (Benzinga)	May 2026	https://app.bigdata.com/documents/84954D11A562DF8249E7187F90B3BCD7
10	Rigetti Computing Company Tearsheet (Bigdata.com)	May 2026	https://bigdata.com
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17	Quantinuum S-1 Filing (SEC)	May 2026	https://www.sec.gov/Archives/edgar/data/2110105/000162828026032836/quantinuum-sx1.htm
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19	IonQ Patent Portfolio (Scope Technologies Report)	Mar 2025	https://app.bigdata.com/documents/350E6B75692DAF1F5A14C904644ADAC6
20	CrispIdeas - Quantum Computing 2025 Review	Feb 2026	https://app.bigdata.com/documents/5A89D0F46FD3081689B71A3EA4F63824
21	QCi Q1 2026 Earnings Call Transcript	May 2026	https://app.bigdata.com/documents/627C146E6DF4D12B1333ABCE966A19C3
22	Fujitsu 2025 Annual Report - Patent Rankings	Oct 2025	https://app.bigdata.com/documents/6273DE619882E1A97FC0F6923D651332
23	Inflection Q1 2026 Earnings Report	May 2026	https://app.bigdata.com/documents/C35595ABDCD1F4CC5E494A19652E52E7
24	IBM Quantum-Assisted Protein Simulation News	May 2026	https://app.bigdata.com/documents/7CC2DEC918AC85A6A84090B8FB1447D1

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